Charles Shaju

Robotics & Embedded Systems Engineer

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SUMMARY

Robotics and embedded engineer with 2+ years in marine robotics (ASVs, underwater ROVs), now pursuing an MCA to move into perception and autonomy. Tuned PID control on an 80 kg catamaran ASV used for dam surveys at 4 sites; building a ROS2 mobile robot with custom Kalman filter sensor fusion on Raspberry Pi and ESP32. Available for a full-time internship from January.

WORK EXPERIENCE

Embedded Systems Engineer (full-time until 06/2025, part-time since)
i4 Marine Technologies Pvt Ltd · Pune, Maharashtra · 08/2023 – Present

- Tuned steering, speed and acceleration gains on an ArduPilot/Pixhawk catamaran ASV (80 kg payload), using Mission Planner gyro graphs and logs to remove path zigzag; held planned survey lines within 1 m during bathymetry and dam-wall inspection at 4 dams.
- Set up UDP telemetry over radio modems and a live sensor-data UI for the ASV up to 2–3 km.
- Integrated pH, turbidity and temperature sensors into the ASV's water-quality payload.
- Wired, configured, calibrated and tested underwater ROVs, including a commercial BlueROV.
- Built IoT speed-control modules for electric canoes; currently building GPS IoT modules for outboard motors and a precision barometer.

EdTech & Robotics Solutions Engineer (Intern)
Jobin & Jismi · Thrissur, Kerala · 04/2026 – 05/2026

- Built a browser-based block-programming platform (Google Blockly + JavaScript) that connects over Bluetooth to an ESP32 Otto robot, so students can drive it and program it with blocks.
- Tested and debugged the educational robot's motor control and movements.

Robotics Intern
Shristi Robotics · Thrissur, Kerala · 01/2023 – 07/2023

- Built a 6-thruster underwater ROV (Pixhawk, ArduSub, 3D-printed components) on a funded project: wiring, calibration and testing; deployed for dam inspection.
- Taught mobile robotics, sensors and Arduino: a 2-week summer course for school students and one-day workshops for college students.

SKILLS

Languages: Python, Embedded C/C++ (Arduino, PlatformIO), JavaScript (Blockly)

Robotics: ROS 2 Humble, RViz, ArduPilot/ArduSub, MAVLink, Mission Planner, QGroundControl, PID tuning, custom Kalman filter sensor fusion, Ubuntu/Linux, Git/GitHub

Hardware: ESP32, Raspberry Pi, Pixhawk, Camera (USB, RealSense D455), 3D printing (FDM), KiCad, OnShape (Basic)

Sensors & Actuators: Barometric sensors, depth, GPS, IMU, motor control, pressure, TFmini

Protocols: I2C, SPI, UART, PWM, LoRa, RS485/Modbus, Wi-Fi, UDP, Bluetooth

EDUCATION

MCA in Computer Application
Rajagiri College of Social Science · Kochi, Kerala · 06/2025 – 04/2027 (expected)

Full-time. Coursework: Data Structures & Algorithms (Python), DBMS, Operating Systems

Bachelor of Vocational Studies (IT)
Christ College Autonomous, Irinjalakuda · Thrissur, Kerala · 08/2020 – 04/2023

Full-time. GPA: 85%

CERTIFICATIONS

- MathWorks Computer Vision Engineer · MathWorks · 07/2026
- Certificate in AI-Machine Learning Developer · ASAP Kerala · 06/2024